1. (例 6.2.9)

$$\begin{aligned} I &= \int \frac{\mathrm{d}x}{\sqrt{x(1-x)}} \\ &= \int \frac{\mathrm{d}x}{\sqrt{\frac{1}{4}-\left(x-\frac{1}{2}\right)^2}} \\ &= \arcsin(2x-1)+C. \end{aligned}$$

2. (例 6.2.11)

$$\begin{aligned} I_n &= \int \tan^n x \mathrm{d}x \\ &= \int \tan^{n-2} x \cdot \frac{1-\cos^2 x}{\cos^2 x} \mathrm{d}x \\ &= \int \tan^{n-2} x \cdot (1-\cos^2 x) \mathrm{d} \tan x \\ &= \frac{\tan^{n-1} x}{n-1} - I_{n-2}. \end{aligned}$$

边界有

$$I_0 = x + C, \quad I_1 = -\ln|\cos x| + C.$$

3. (习题六 3)

(2)

$$\begin{aligned} \int \frac{\mathrm{d}x}{\sqrt{2-x^2}} &= \int \frac{\mathrm{d}\frac{x}{\sqrt{2}}}{\sqrt{1-\left(\frac{x}{\sqrt{2}}\right)^2}} \\ &= \arcsin \frac{x}{\sqrt{2}} + C. \end{aligned}$$

(4)

$$\begin{aligned} \int \frac{\mathrm{d}x}{1+\sin x} &= \int \frac{1-\sin x}{\cos^2 x} \mathrm{d}x \\ &= \int \frac{1}{\cos^2 x} \mathrm{d}x - \int \frac{\sin x}{\cos^2 x} \mathrm{d}x \\ &= \tan x - \sec x + C. \end{aligned}$$

(6)

$$\begin{aligned} \int \frac{x\mathrm{d}x}{\sqrt{1-x^2}} &= \frac{1}{2} \int \frac{\mathrm{d}x^2}{\sqrt{1-x^2}} \\ &= -\sqrt{1-x^2} + C. \end{aligned}$$

(8)

$$\begin{aligned}\int \frac{e^x}{1+e^x}\mathrm{d}x &= \int \frac{\mathrm{d}e^x}{1+e^x} \\ &= \ln(1+e^x)+C.\end{aligned}$$

(10)

$$\begin{aligned}\int \frac{\mathrm{d}x}{x\sqrt{x^2-1}} &= \frac{1}{2}\int \frac{\mathrm{d}x^2}{x^2\sqrt{x^2-1}} \\ &= \frac{1}{2}\int \frac{\mathrm{d}\sqrt{x^2-1}^2}{\left(\sqrt{x^2-1}^2+1\right)\sqrt{x^2-1}} \\ &= \int \frac{\mathrm{d}\sqrt{x^2-1}}{1+\sqrt{x^2-1}} \\ &= \arctan\sqrt{x^2-1}+C.\end{aligned}$$

(12)

$$\begin{aligned}I := \int \sin^2 x\cos^2 x\mathrm{d}x &= \frac{1}{4}\int \sin^2 2x\mathrm{d}x \\ &= -\frac{1}{8}\int \sin 2x\cdot \mathrm{d}\cos 2x \\ &= -\frac{1}{8}\left(\sin 2x\cos 2x - 2\int \cos^2 2x\cdot \mathrm{d}x\right) \\ &= -\frac{1}{8}(\sin 2x\cos 2x - 2x + 8I) \\ &= -\frac{1}{16}(\sin 2x\cos 2x - 2x)+C.\end{aligned}$$

(14)

$$\begin{aligned}\int \frac{\sin x\cos x}{\sqrt{a^2\sin^2 x+b^2\cos^2 x}}\mathrm{d}x &= \int \frac{\sin x\cos x}{a\sqrt{1-e^2\cos^2 x}}\mathrm{d}x \qquad \left(e^2=\frac{a^2-b^2}{a^2}\right) \\ &= -\frac{1}{a}\int \frac{\cos x}{\sqrt{1-e^2\cos^2 x}}\mathrm{d}\cos x \\ &= -\frac{1}{2a}\int \frac{\mathrm{d}t}{\sqrt{1-e^2t}} \qquad (t=\cos^2 x) \\ &= -\frac{1}{2a}\cdot -\frac{2}{e^2}\sqrt{1-e^2t} \\ &= \frac{1}{a^2-b^2}\sqrt{a^2-(a^2-b^2)\cos^2 x} \\ &= \frac{1}{a^2-b^2}\sqrt{a^2\sin^2 x+b^2\cos^2 x}+C.\end{aligned}$$

(16)

$$\begin{aligned}\int \frac{\mathrm{d}x}{x^2\sqrt{1+x^2}} &= \int \frac{(x^2+1)-x^2}{x^2\sqrt{1+x^2}}\mathrm{d}x \\ &= \int \frac{(1+x^2)^{1/2}-x^2(1+x^2)^{-1/2}}{x^2}\mathrm{d}x \\ &= -\frac{\sqrt{x^2+1}}{x}+C.\end{aligned}$$

(18)

$$
\begin{aligned}
\int \frac{x}{1+\sqrt{x}}\mathrm{d}x &= \int \frac{(u-1)^2}{u}\mathrm{d}(u-1)^2 \quad \left(u=1+\sqrt{x}\right) \\
&= 2\int \frac{u^3-3u^2+3u-1}{u}\mathrm{d}u \\
&= 2\left(\frac{u^3}{3}-\frac{3u^2}{2}+3u-\ln|u|\right)+C \\
&= \frac{2}{3}(1+\sqrt{x})^3-3(1+\sqrt{x})^2+6(1+\sqrt{x})-2\ln(1+\sqrt{x})+C \\
&= \frac{2}{3}x^{3/2}-x+2\sqrt{x}-2\ln(1+\sqrt{x})+C.
\end{aligned}
$$

(20)

$$
\begin{aligned}
\int \frac{\mathrm{d}x}{x\sqrt{4-x^2}} &= \frac{1}{2}\int \frac{\mathrm{d}x^2}{x^2\sqrt{4-x^2}} \\
&= \frac{1}{2}\int \frac{-\mathrm{d}u^2}{(4-u^2)u} \quad \left(u=\sqrt{4-x^2}\right) \\
&= \int \frac{\mathrm{d}u}{u^2-4} \\
&= \frac{1}{4}\left(\int \frac{\mathrm{d}u}{u-2}-\int \frac{\mathrm{d}u}{u+2}\right) \\
&= \frac{1}{4}\ln\left|\sqrt{4-x^2}-2\right|-\frac{1}{4}\ln\left|\sqrt{4-x^2}+2\right|+C.
\end{aligned}
$$

---

4. (例 6.2.22)

$$
\begin{aligned}
I &= \int e^{ax}\cos bx\mathrm{d}x \\
&= \frac{1}{b}\int e^{ax}\mathrm{d}(\sin bx) \\
&= \frac{1}{b}e^{ax}\sin bx-\frac{1}{b}\int \sin bx\cdot ae^{ax}\mathrm{d}x \\
&= \frac{1}{b}e^{ax}\sin bx-\frac{a}{b}J.
\end{aligned}
$$

同理,

$$
\begin{aligned}
J &= \int e^{ax}\sin bx\mathrm{d}x \\
&= -\frac{1}{b}\int e^{ax}\mathrm{d}(\cos bx) \\
&= -\frac{1}{b}e^{ax}\cos bx+\frac{1}{b}\int \cos bx\cdot ae^{ax}\mathrm{d}x \\
&= -\frac{1}{b}e^{ax}\cos bx+\frac{a}{b}I.
\end{aligned}
$$

解得

$$
\begin{aligned}
I &= \frac{e^{ax}(b\sin bx+a\cos bx)}{a^2+b^2}+C_1, \\
J &= \frac{e^{ax}(a\sin bx-b\cos bx)}{a^2+b^2}+C_2.
\end{aligned}
$$

5. (例 6.2.23)

注意到:

$$bI + aJ = x + C_1,$$

以及

$$\begin{aligned} bJ - aI &= \int \frac{b\cos x - a\sin x}{a\cos x + b\sin x}\mathrm{d}x \\ &= \int \frac{\mathrm{d}(a\sin x + b\cos x)}{a\cos x + b\sin x} \\ &= \ln|a\cos x + b\sin x| + C_2. \end{aligned}$$

解得

$$\begin{aligned} I &= \frac{bx - a\ln|a\cos x + b\sin x|}{a^2+b^2} + C_3, \\ J &= \frac{ax + b\ln|a\cos x + b\sin x|}{a^2+b^2} + C_4. \end{aligned}$$

6. (例 6.2.25)

$$\begin{aligned} I_n &= \int \sin^n x\mathrm{d}x \\ &= -\int \sin^{n-1} x\mathrm{d}\cos x \\ &= -\sin^{n-1} x\cos x + (n-1)\int \cos x\sin^{n-2} x\mathrm{d}(\sin x) \\ &= -\sin^{n-1} x\cos x + (n-1)\int (1-\sin^2 x)\sin^{n-2} x\mathrm{d}x \\ &= \frac{n-1}{n}I_{n-2} - \frac{1}{n}\sin^{n-1} x\cos x. \end{aligned}$$

边界有

$$I_0 = x + C_0, \quad I_1 = -\cos x + C_1.$$

7. (习题六 4)

(2)

$$\begin{aligned} \int xe^{-3x}\mathrm{d}x &= -\frac{1}{3}\int x\mathrm{d}e^{-3x} \\ &= -\frac{1}{3}xe^{-3x} + \frac{1}{3}\int e^{-3x}\mathrm{d}x \\ &= -\frac{1}{3}xe^{-3x} - \frac{1}{9}e^{-3x} + C. \end{aligned}$$

(4)

$$
\begin{aligned}
\int \arcsin x \mathrm{d}x &= x\arcsin x - \int \frac{x}{\sqrt{1-x^2}}\mathrm{d}x \\
&= x\arcsin x - \frac{1}{2}\int \frac{\mathrm{d}x^2}{\sqrt{1-x^2}} \\
&= x\arcsin x + \sqrt{1-x^2} + C.
\end{aligned}
$$

(6)

$$
\begin{aligned}
I := \int \frac{\arctan x}{x^2(1+x^2)}\mathrm{d}x &= \underbrace{\int \frac{\arctan x}{x^2}\mathrm{d}x}_{J} - \int \arctan x \mathrm{d}\arctan x \\
&= J - \frac{1}{2}\arctan^2 x.
\end{aligned}
$$

再有

$$
\begin{aligned}
J &= \frac{\arctan x}{x} - \int x\frac{\frac{x^2}{1+x^2} - 2x\arctan x}{x^4}\mathrm{d}x \\
&= \frac{\arctan x}{x} - \frac{1}{2}\int \frac{\mathrm{d}x^2}{x^2(1+x^2)} + 2\int \frac{\arctan x}{x^2}\mathrm{d}x \\
&= \frac{\arctan x}{x} - \frac{1}{2}\left(\int \frac{\mathrm{d}x^2}{x^2} - \int \frac{\mathrm{d}x^2}{1+x^2}\right) + 2J \\
&= \frac{1}{2}\ln\frac{x^2}{1+x^2} - \frac{\arctan x}{x} + C.
\end{aligned}
$$

最终

$$
I = \frac{1}{2}\ln\frac{x^2}{1+x^2} - \frac{\arctan x}{x} - \frac{1}{2}\arctan^2 x + C.
$$

(8)

$$
\begin{aligned}
\int \frac{x^2}{(1+x^2)^2}\mathrm{d}x &= \int \frac{1}{1+x^2}\mathrm{d}x - \int \frac{1}{(1+x^2)^2}\mathrm{d}x \\
&= \arctan x - \int \frac{1}{(1+x^2)^2}\mathrm{d}x \\
&= \arctan x + \frac{1}{2}\int \frac{1}{x}\mathrm{d}\frac{1}{1+x^2} \\
&= \arctan x + \frac{1}{2}\left(\frac{1}{x(1+x^2)} + \int \frac{1}{(1+x^2)x^2}\mathrm{d}x\right) \\
&= \arctan x + \frac{1}{2x(1+x^2)} + \frac{1}{2}\left(\int \frac{1}{x^2}\mathrm{d}x - \int \frac{1}{1+x^2}\mathrm{d}x\right) \\
&= \frac{1}{2}\arctan x - \frac{x}{2(1+x^2)} + C.
\end{aligned}
$$

(10)

$$
\begin{aligned}
I := \int e^x \cos^2 x \mathrm{d}x &= \int e^x \cos x \mathrm{d}(\sin x) \\
&= e^x \cos x \sin x - \int \sin x \mathrm{d}(e^x \cos x) \\
&= e^x \cos x \sin x - \int e^x \sin x (\cos x - \sin x) \mathrm{d}x \\
&= e^x \cos x \sin x - \underbrace{\int e^x \sin x \cos x \mathrm{d}x}_{J} + \underbrace{\int e^x \sin^2 x \mathrm{d}x}_{K}.
\end{aligned}
$$

注意到

$$
I + K = \int e^x \mathrm{d}x = e^x + C,
$$

接下来只需要求出 J.

$$
\begin{aligned}
J &= \int e^x \sin x \mathrm{d} \sin x \\
&= e^x \sin^2 x - \int e^x \sin x (\sin x + \cos x) \mathrm{d}x \\
&= e^x \sin^2 x - K - J + C.
\end{aligned}
$$

因此

$$
\begin{cases}
I + J - K = e^x \cos x \sin x + C_1, \\
I + K = e^x + C_2, \\
2J + K = e^x \sin^2 x + C_3.
\end{cases}
$$

解得

$$
I = \frac{e^x}{5} \left(\sin 2x + \cos^2 x + 2 \right) + C.
$$

Supplementary.

> **S1.**
>
> 计算不定积分:
>
> (1) $\displaystyle\int x^3 \sqrt[3]{1 + x^2} \mathrm{d}x$;
>
> (2) $\displaystyle\int \sqrt{1 + x^2} \mathrm{d}x$;
>
> (3) $\displaystyle\int x^2 \sqrt{1 + x^2} \mathrm{d}x$;
>
> (4) $\displaystyle\int \frac{\mathrm{d}x}{1 + \sqrt{x} + \sqrt{x+1}}$;
>
> (5) $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{(b-x)(x-a)}}$;
>
> (6) $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{x^2 + x}}$;
>
> (7) $\displaystyle\int \frac{x^3}{\sqrt{x^4 - 2x^2 - 1}} \mathrm{d}x$.

$$
\begin{aligned}
(1) &= \int x^3(1+x^2)^{\frac{1}{3}}\mathrm{d}x \\
&= \frac{1}{2}\int x^2(1+x^2)^{\frac{1}{3}}\mathrm{d}x^2 \\
&= \frac{1}{2}\int (u^3-1)u\cdot \mathrm{d}u^3 \quad (u=\sqrt[3]{1+x^2}) \\
&= \cdots
\end{aligned}
$$

$$
\begin{aligned}
(2) &= x\sqrt{x^2+1} - \int \frac{(x^2+1)-1}{\sqrt{1+x^2}}\mathrm{d}x \\
&= x\sqrt{x^2+1} - \int \sqrt{x^2+1}\mathrm{d}x + \int \frac{\mathrm{d}x}{\sqrt{1+x^2}} \\
&= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\int \frac{\mathrm{d}\sinh u}{\cosh u} \quad (x=\sinh u) \\
&= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}u + C \\
&= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\ln(x+\sqrt{1+x^2}) + C.
\end{aligned}
$$

$$
\begin{aligned}
(3) &= \frac{1}{3}\int x\mathrm{d}(1+x^2)^{\frac{3}{2}} \\
&= \frac{1}{3}x(1+x^2)^{\frac{3}{2}} - \frac{1}{3}\int \sqrt{1+x^2}\mathrm{d}x - \frac{1}{3}\int x^2\sqrt{1+x^2}\mathrm{d}x \\
&= \cdots
\end{aligned}
$$

(4) 见 作业十三 6.(2), 也可以用 (2) 的结论.

$$
\begin{aligned}
(5) &= \int \frac{\mathrm{d}x}{\sqrt{\left(\frac{b-a}{2}\right)^2 - \left(\frac{b+a}{2}-x\right)^2}} \\
&= -\arcsin\frac{\frac{b+a}{2}-x}{\frac{b-a}{2}} + C.
\end{aligned}
$$

(6) 也是在分母配方, 最后的形式类似一个双曲正弦的反函数.

$$
\begin{aligned}
(7) &= \frac{1}{2}\int \frac{t\mathrm{d}t}{\sqrt{t^2-2t-1}} \\
&= \frac{1}{2}\int \frac{\frac{1}{2}\mathrm{d}(t^2-2t-1)}{\sqrt{t^2-2t-1}} + \int \frac{\mathrm{d}t}{\sqrt{t^2-2r-1}}.
\end{aligned}
$$

后者仍是配方.

> **S2.**
>
> (1) $\displaystyle\int \frac{\mathrm{d}x}{1-\cos x}$;
>
> (2) $\displaystyle\int \tan^3 x\mathrm{d}x$;
>
> (3) $\displaystyle\int \sin^4 x\mathrm{d}x$;
>
> (4) $\displaystyle\int \frac{\mathrm{d}x}{\sin x}$;

(5) $\displaystyle\int \frac{\mathrm{d}x}{\sin^4 x}$;

(6) $\displaystyle\int \frac{x\tan x}{\cos^2 x}\mathrm{d}x$;

(7) $\displaystyle\int \frac{x\arctan x}{(1+x^2)^2}\mathrm{d}x$.

$$
\begin{aligned}
(1) &= \int \frac{\mathrm{d}x}{2\sin^2\frac{x}{2}} \\
&= -\cot\frac{x}{2} + C.
\end{aligned}
$$

$$
\begin{aligned}
(2) &= \int \frac{\sin^3 x}{\cos^3 x}\mathrm{d}x \\
&= \int \frac{\cos^2 x - 1}{\cos^3 x}\mathrm{d}\cos x \\
&= \cdots
\end{aligned}
$$

$$
\begin{aligned}
(3) &= \int \left(\frac{1-\cos 2x}{2}\right)^2 \mathrm{d}x \\
&= \frac{1}{4}\int (1 - 2\cos 2x + \cos^2 2x)\mathrm{d}x \\
&= \cdots
\end{aligned}
$$

$$
\begin{aligned}
(4) &= \int \frac{\sin x}{\sin^2 x}\mathrm{d}x \\
&= -\int \frac{\mathrm{d}\cos x}{1-\cos^2} \\
&= \cdots
\end{aligned}
$$

$$
(5) = \int (1+u^2)\mathrm{d}u \quad (u = \cot x).
$$

$$
\begin{aligned}
(6) &= \int x\mathrm{d}\left(\frac{1}{2}\tan^2 x\right) \\
&= \frac{x}{2}\tan^2 x - \frac{1}{2}\int \frac{1-\cos^2 x}{\cos^2 x}\ x \\
&= \frac{x}{2}\tan^2 x - \frac{1}{2}\tan x + \frac{x}{2} + C.
\end{aligned}
$$

$$
\begin{aligned}
(7) &= -\frac{1}{2}\int \arctan x \cdot \mathrm{d}\frac{1}{1+x^2} \\
&= - = \frac{1}{2}\frac{\arctan x}{1+x^2} + \frac{1}{2}\int \frac{\mathrm{d}x}{(1+x^2)^2}.
\end{aligned}
$$

参考 S5 (1).

S3.

$$
(1) = \int \frac{1}{1+u}\frac{\mathrm{d}u}{u} \quad (u = e^x).
$$

$$
(2) = \int \frac{1}{u}\frac{u^2+1}{u}\mathrm{d}u \quad (u = \sqrt{e^x - 1}).
$$

$$(3)=\int\left(e^x-4\left(\frac{e^x}{x}-\frac{e^x}{x^2}\right)\right)\mathrm{d}x,$$

其中

$$\int\frac{e^x}{x}\mathrm{d}x=\int\frac{\mathrm{d}e^x}{x}=\frac{e^x}{x}+\int\frac{e^x}{x^2}\mathrm{d}x.$$

> **S5.**
>
> (1) $I_n=\int(1+x^2)^{-n}\mathrm{d}x$;
>
> (2) $I_n=\int\frac{\sin nx}{\sin x}\mathrm{d}x$;

(1)

$$\begin{aligned}I_n&=\frac{x}{(1+x^2)^n}+2n\int\frac{(1+x^2)-1}{(1+x^2)^{n+1}}\mathrm{d}x\\&=\frac{x}{(1+x^2)^n}+2nI_n-2nI_{n+1}.\end{aligned}$$

(2) $I_1=x+C$, $I_2=2\sin x+C$.

$$\begin{aligned}I_{n+2}-I_n&=\int\frac{\sin(n+2)x-\sin nx}{\sin x}\mathrm{d}x\\&=2\int\cos(n+1)x\mathrm{d}x\\&=\frac{2}{n+1}\sin(n+1)x\mathrm{d}x.\end{aligned}$$

> **S6.**
>
> (1) $I=\int\frac{\sin x}{2\sin x+3\cos x}\mathrm{d}x$;
>
> (2) $I=\int\frac{\mathrm{d}x}{1+x^4}$;
>
> (3) $I=\frac{\mathrm{d}x}{x^8+x^4+1}$.

(1) 令

$$J=\int\frac{\cos x}{2\sin x+3\cos x}\mathrm{d}x,$$

那么

$$2I+3J=xx+C;$$
$$2J-3I=\int\frac{\mathrm{d}(2\sin x+3\cos x)}{2\sin x+3\cos x}=\ln|2\sin x+3\cos x|+C.$$

即可解出 I.

(2) 令

$$J=\int\frac{x^2\mathrm{d}x}{1+x^4},$$

那么

$$\begin{aligned}I+J&=\int\frac{1+x^2}{1+x^4}\mathrm{d}x\\&=\int\frac{1+x^{-2}}{x^{-2}+x^2}\mathrm{d}x\\&=\int\frac{\mathrm{d}\left(x-\frac{1}{x}\right)}{\left(x-\frac{1}{x}\right)^2+2}\\&=\frac{1}{\sqrt{2}}\arctan\frac{x+\frac{1}{x}}{\sqrt{2}}.\end{aligned}$$

类似地研究出 $I-J$ 即可.

(3) 注意

$$x^8+x^4+1=(x^4+1)^2-x^4=(x^4+x^2+1)(x^4-x^2+1).$$

考虑凑 $1\pm(x^2)^3$ 的形式, 有

$$\frac{1}{x^8+x^4+1}=\frac{1}{2}\left(\frac{1+x^2}{x^4+x^2+1}+\frac{1-x^2}{x^4+x^2+1}\right).$$

这就可以类似 (2) 地处理了.